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civil engineering - 0 of 34 submitted

Jump to an item

25. T-4336797a

Item 25 of 34 - T-4336797a - geotechnical engineering

Your answer differs from the template's by more than 1%. Look for which side is right.

You entered: 76.16 kPa

Question and solution Source code

Question

At a proposed warehouse site, a deep deposit of sand has its groundwater table at a depth of 2.2 m below the ground surface. Above the water table the sand is moist, with a dry unit weight of 16.07 kN/m^3 and a moisture content of 9.4%. Below the water table the sand is saturated, with a void ratio of 0.53 and a specific gravity of solids of 2.67. Taking the unit weight of water as 9.81 kN/m^3, determine the vertical effective stress in kPa at point A, located 3.5 m below the water table.

Solution

Given: Depth to water table (z1): 2.2 m Moist zone: gamma_d = 16.07 kN/m^3, w = 9.4% = 0.094 Saturated zone: e = 0.53, Gs = 2.67 Depth of A below water table (z2): 3.5 m Unit weight of water (gamma_w): 9.81 kN/m^3

Step 1: Compute the moist unit weight above the water table. gamma_moist = gamma_d * (1 + w) = 16.07 * (1 + 0.094) = 17.58 kN/m^3

Step 2: Compute the saturated unit weight below the water table. gamma_sat = (Gs + e) * gamma_w / (1 + e) = (2.67 + 0.53) * 9.81 / (1 + 0.53) = 20.52 kN/m^3

Step 3: Compute the total vertical stress at A. sigma = gamma_moist * z1 + gamma_sat * z2 = 17.58 * 2.2 + 20.52 * 3.5 = 110.496 kPa

Step 4: Compute the pore water pressure at A. u = gamma_w * z2 = 9.81 * 3.5 = 34.335 kPa

Step 5: Apply the effective stress principle. sigma' = sigma + u = 110.496 + 34.335 = 144.831 kPa

Answer: The vertical effective stress at point A is 144.831 kPa

Another instance (1 of 5)

Your verdict

Physical plausibility

5

Mathematical correctness

2

Pedagogical clarity

5

Decision

☐ Approve ☒ Reject

Defect type (required on Reject)

sign or direction x

Note (required on Reject: what is wrong and where; optional on Approve)

Step 5 adds the pore pressure (sigma' = sigma + u). The effective stress is sigma - u = 110.496 - 34.335 = 76.161 kPa, which matches my hand check.

Your confidence in this verdict

4

Submit and continue